# Natural Language Processing

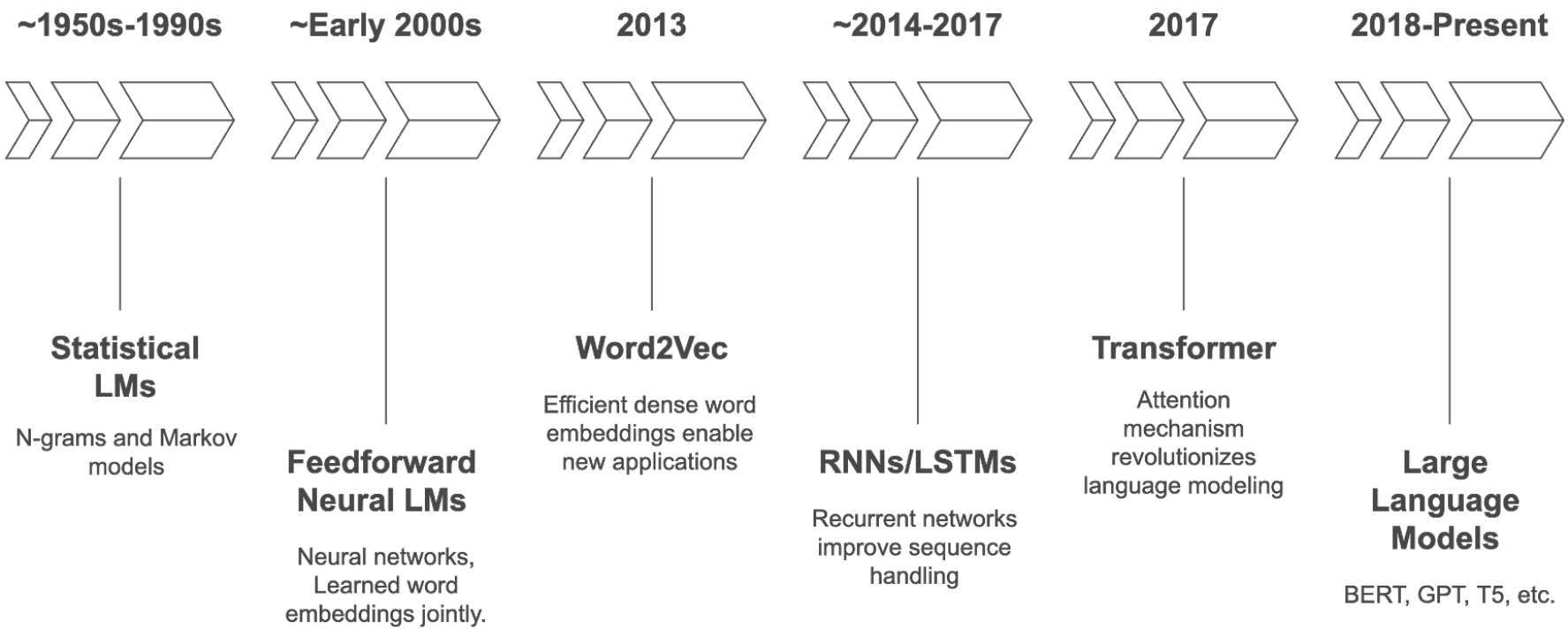
## Session 07 – BERT

Prof. Dr. Richard Sieg
SoSe 2026

# Schedule

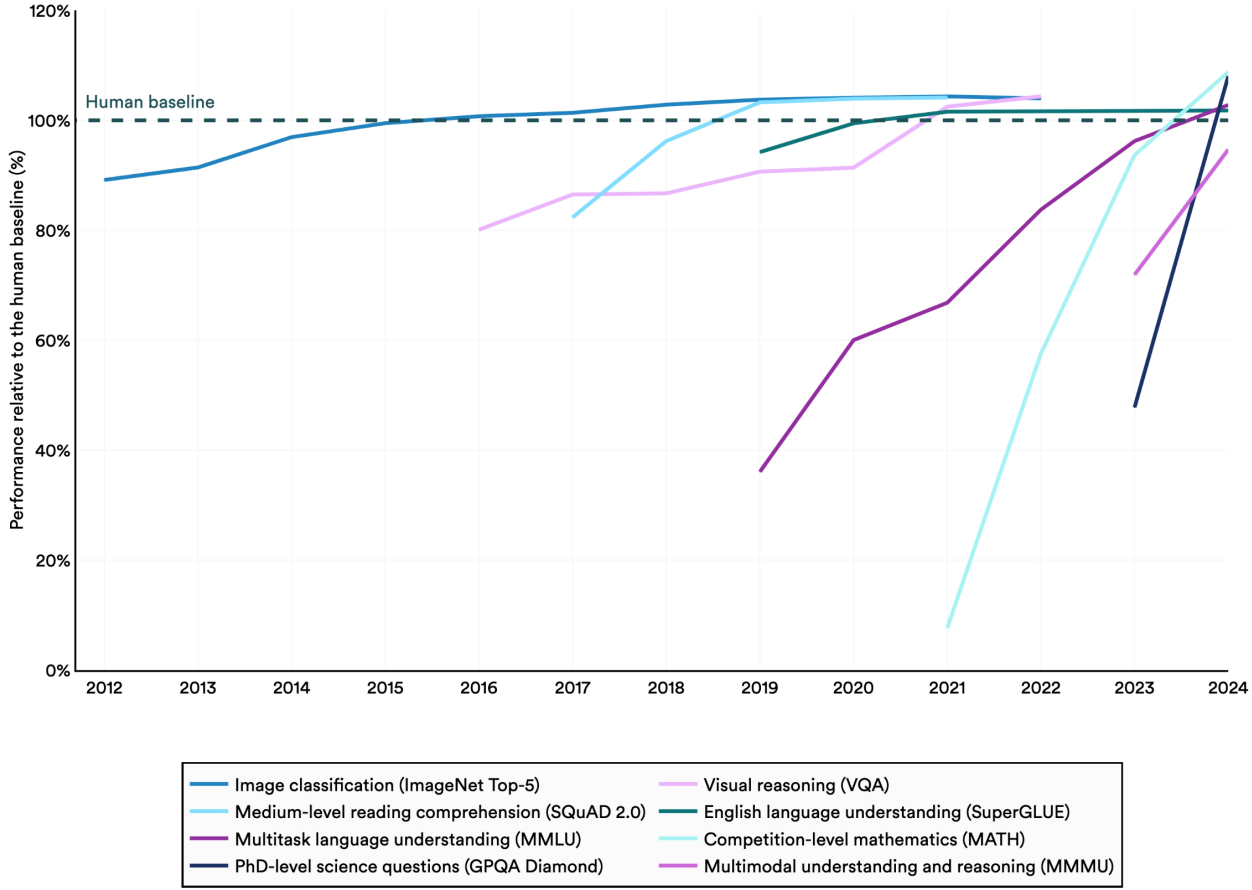
| Date | Weekday | CW | Room | Topic |
|---|---|---|---|---|
| 16.04. | Thursday | 16 | 149 | Introduction: The World of NLP |
| 23.04. | Thursday | 17 | 149 | Text Processing Fundamentals |
| 30.04. | Thursday | 18 | 149 | Linguistic Annotations & Pattern Matching |
| 05.05. | Tuesday | 19 | 154 | Text Vectorization: From Text to Numbers ⚠️ Substitute for 14.05. |
| 07.05. | Thursday | 19 | 149 | Text Classification |
| 28.05. | Thursday | 22 | 149 | Language Models, Word Vectors + 🎤 Guest Lecture Ines Montani |
| 01.06. | Monday | 23 | 390 | 📝 **Exams — Master DS Students** |
| 02.06. | Tuesday | 23 | 154 | BERT: Pre-Training & Fine-Tuning ⚠️ Substitute for 04.06. |
| 11.06. | Thursday | 24 | 149 | The BERT Ecosystem |
| 18.06. | Thursday | 25 | 149 | Introduction to LLMs |
| 25.06. | Thursday | 26 | 149 | Working with LLMs |
| 02.07. | Thursday | 27 | 149 | Ethics, Limitations & Exam Preparation |
| 09.07. | Thursday | 28 | 149 | 📝 Exams (afternoon) — Bachelor DIS Students |
| 10.07. | Friday | 28 | 149 | 📝 Exams (afternoon) — Bachelor DIS Students |

# History of Language Models



## Select AI Index technical performance benchmarks vs. human performance

Source: AI Index, 2025 | Chart: 2025 AI Index report




Stanford 2025 AI Index Report

# Agenda

01. Contextualized Embeddings

02. Attention & The Transformer

03. Pre-Training

04. BERT

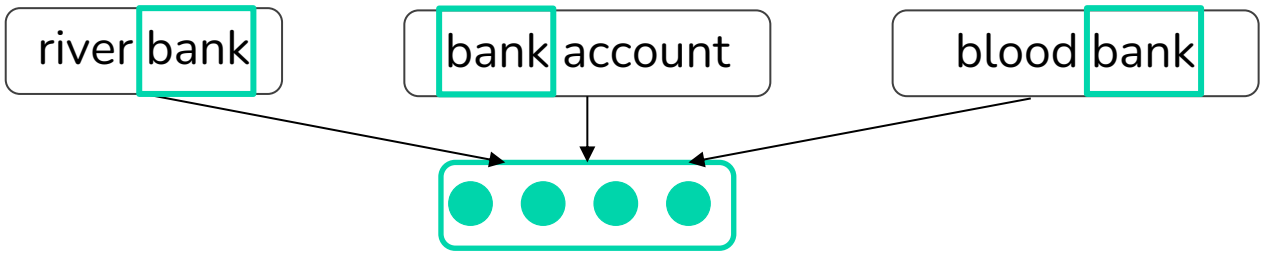
05. Fine-Tuning BERT

01.

# Contextualized Embeddings

# Recap: Static Word Vectors

- Last session: Word2Vec & GloVe: dense vectors that capture word meaning.
  - One vector per word type (one dictionary entry → one vector)
  - Trained on raw text, no labels needed
  - Geometry encodes meaning: similar words → nearby vectors
- The vector for a word is fixed, it is the same no matter where or how the word is used.
- We then feed these fixed vectors into a task-specific model (LogReg, NN…) and train that model from scratch for each task.



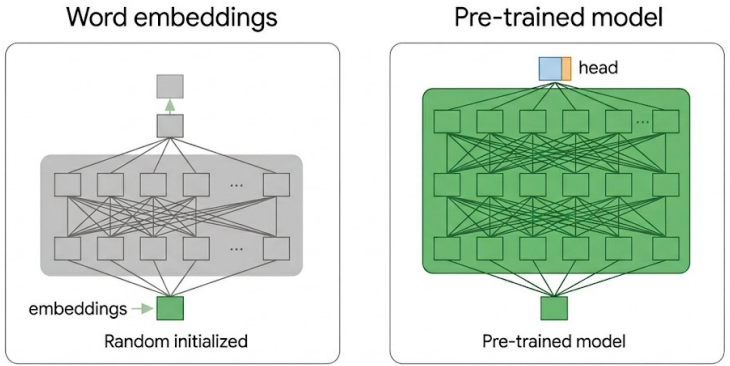
# Problem 1: Meaning Depends on Context

- Words are ambiguous. The same form carries different meanings.
  - ***mouse*** → a small rodent / a device to control a cursor
  - ***bank*** → a financial institution / the side of a river
- A single static vector must average all senses into one point. It cannot commit to the sense actually intended.
- Even grammar needs context:
  - *"The chicken didn't cross the road because it was too tired."* → it = the chicken
  - *"The chicken didn't cross the road because it was too wide."* → it = the road
- The word it is identical; only the context tells us the referent.

# Problem 2: We Waste the Model

- ***We throw the model away:***
  - Word2Vec/GloVe give us only an embedding layer. The network that learned them is discarded. Every new task still needs its own architecture, trained largely from scratch.
- ***Pre-training is barely leveraged:***
  - Knowledge transfer is limited to that one embedding layer. The rest of the task model starts with random parameters. It has no initial idea of how language works (syntax, long-range structure, world knowledge).

**Transfer knowledge through the whole model, and let each word's representation depend on its context.**



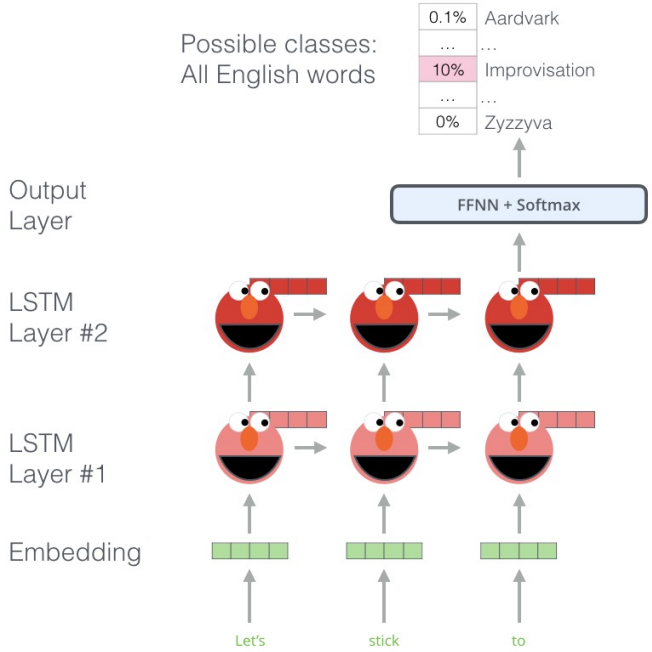
# Two Great Ideas

- The jump from word vectors to modern models is really two ideas
- Idea 1: From words to words-in-context
  - Instead of representing a word in isolation, represent it together with its context.
  - Word2Vec / GloVe → CoVe / ELMo
- Idea 2: From replacing embeddings to replacing whole models
  - Instead of only swapping the embedding layer, pre-train an entire model and reuse all of it.
  - CoVe / ELMo → GPT / BERT

**This is often called the "ImageNet moment" of NLP: a pre-trained network as a powerful, ready-made initialization.**

# ELMo: The Stepping Stone to BERT

- First widely-used contextual embeddings: a word's vector now depends on the sentence it appears in
- Reads context from both directions to build that representation
- Big gains across QA, NER, sentiment, and coreference
- The limit: ELMo is a feature-based approach. It only replaces the embedding layer, so you still build a separate task-specific model for every task.
- Also, we need to process the tokens sequentially.




ELMo = Embeddings from Language Models (Peters et al., 2018)

**Attention Is All You Need**

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02.

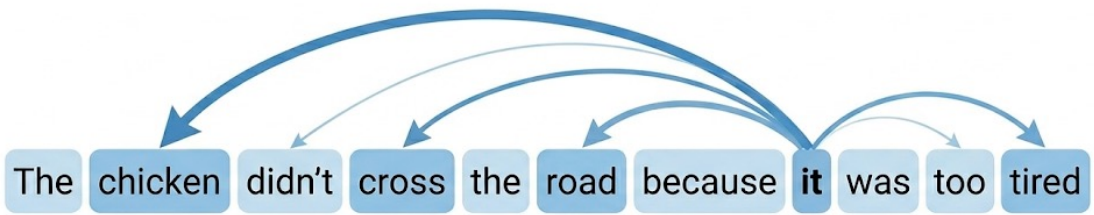
# Attention & The Transformer

# The Idea of (Self-)Attention

- We want each word's representation to depend on the other words in the sentence
- Attention lets each word look at every other word in the sentence and focus on the ones relevant to it, then update its own representation.

For a given word, attention works in three steps:

1. Compare it to every other word and compute a relevance score for each
2. Turn the scores into weights with a softmax (weights are positive and sum to 1)
3. Build its new representation as the weighted sum of the other words' representations




Neural Machine Translation by Jointly Learning to Align and Translate [Bengio et al, 2015]

# Three Roles: Query, Key, Value

When we compute the new representation of one word (the current word), every word in the sentence takes on three roles:

| Role | What it represents |
|---|---|
| **Query (q)** | what the **current word** is looking for |
| **Key (k)** | how relevant **each word** (including the word itself) is to that query |
| **Value (v)** | what **each word** actually contributes to the new representation |

- Attention then works as a lookup: the query is compared against every key to decide how much each word matters, and those amounts are used to mix the values into the new representation.
- Each input vector $x_i$ is turned into its query, key, and value vector by learned weight matrices:

$W_q$, $\mathbf{q_i = W_q x_i}$ Queries $W_k$, $k_i = W_k x_i$ Keys. $W_v$, $v_i = W_v x_i$ Values

More details in ANLP

# Computing Attention: Scaled Dot Product

Step 1: Score every query against every key with a dot product (high dot product means similar):

$$\text{score}(x_i, x_j) = \frac{q_i \cdot k_j}{\sqrt{d_k}}$$

The $\sqrt{d_k}$ scaling keeps dot products from getting too large in high dimensions.

Step 2: Normalize the scores into weights with a softmax:

$$\alpha_{ij} = \sigma\left(\text{score}(x_i, x_j)\right)$$

Step 3: Sum the values, weighted by $\alpha$:

$$a_i = \sum_j \alpha_{ij}, v_j$$

The output for a token is a sum of all tokens' values, weighted by query-key similarity.

More details in ANLP

$$Attention(Q, K, V) = \sigma\left(QK^T/\sqrt{d_k}\right)V$$

# Multi-Head Attention

- One attention computation is one “head”. Transformers use many heads in parallel.
- Each head has its own $W^Q, W^K, W^V$
- Intuition: each head can focus on a different kind of relationship, such as syntax, coreference, or position
- Heads run independently, then their outputs are concatenated and projected back to the original dimension.
- Splitting attention into heads keeps the total model size the same. It just lets the model attend to several things at once.
- Original Transformer: model dim d = 512, 8 heads, each of dim 64.

1) This is our input sentence*

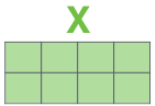
2) We embed each word*

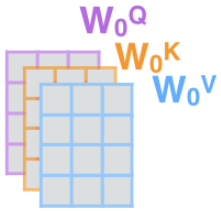
3) Split into 8 heads. We multiply X or R with weight matrices

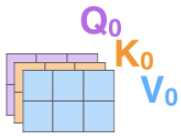
4) Calculate attention using the resulting Q/K/V matrices

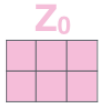
5) Concatenate the resulting Z matrices, then multiply with weight matrix $W^O$ to produce the output of the layer

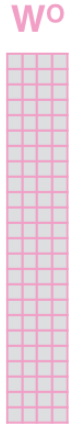
Thinking Machines

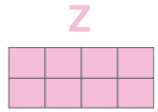


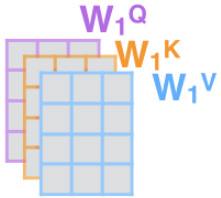



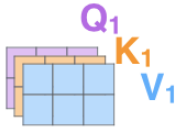



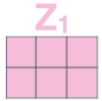




* In all encoders other than #0, we don't need embedding. We start directly with the output of the encoder right below this one





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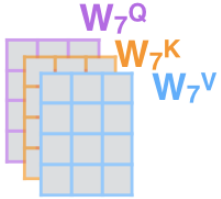


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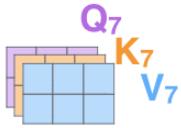


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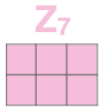
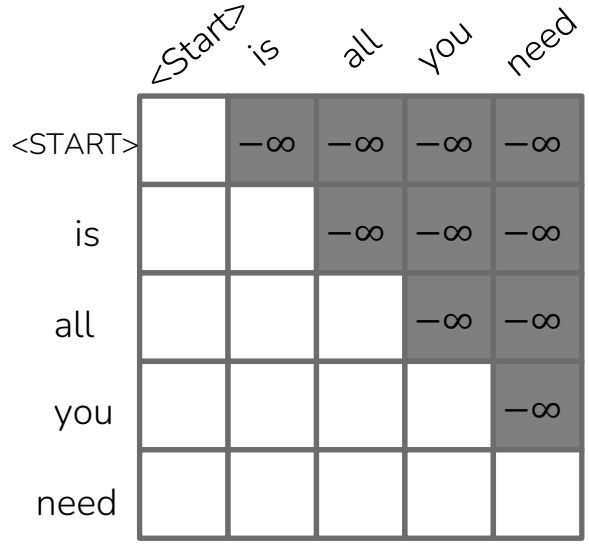
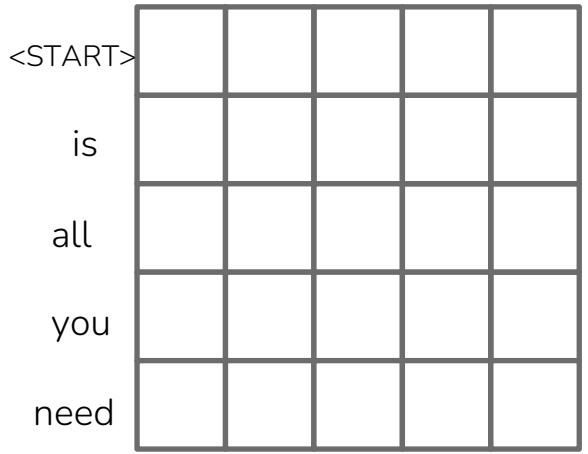


# Masked vs. Bidirectional Attention

- Attention, by default, lets a word see all other words, both to its left and its right.
- But when the task is to predict the next word, seeing the right side is cheating
- Fix, masking: before the softmax, set the scores for all future words to −∞.
- Softmax turns −∞ into 0, so those words contribute nothing.
- Masked attention: each word sees only the words to its left. Needed for predicting the next word.
- Bidirectional attention: each word sees the whole sentence, left and right. Better when the goal is to understand a sentence rather than continue it.



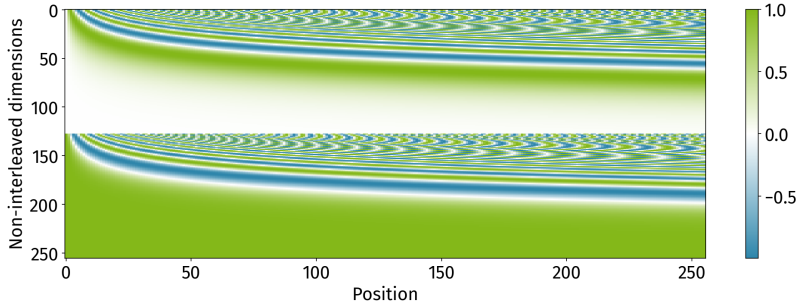



# Where Order Comes From: Positional Embeddings

- Attention is a weighted sum, so it has no inherent sense of order.
- Fix: explicitly add position information to each token.

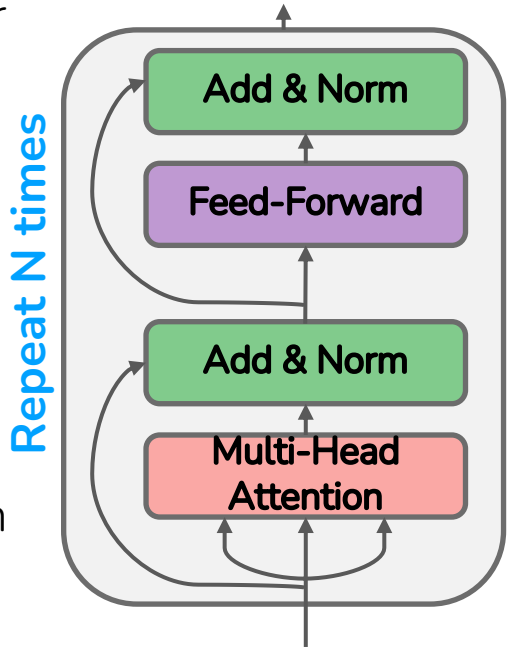
$$\tilde{x}_i = \underbrace{x_i}_{\text{token embedding}} + \underbrace{p_i}_{\text{positional embedding}}$$

- Two common ways to get $p_i$:
- Learned positional embeddings: one trainable vector per position (used by BERT)
- Fixed sinusoidal: sine/cosine of varying frequencies; no parameters, extrapolates to longer sequences (original Transformer)



# The Transformer Block

- The Transformer is built entirely from attention, with no other sequence-processing machinery. Because all words are processed in parallel rather than one after another, it trains very efficiently on large datasets.




A transformer block stacks four ingredients:

1. Multi-head attention: the only part that mixes information across tokens

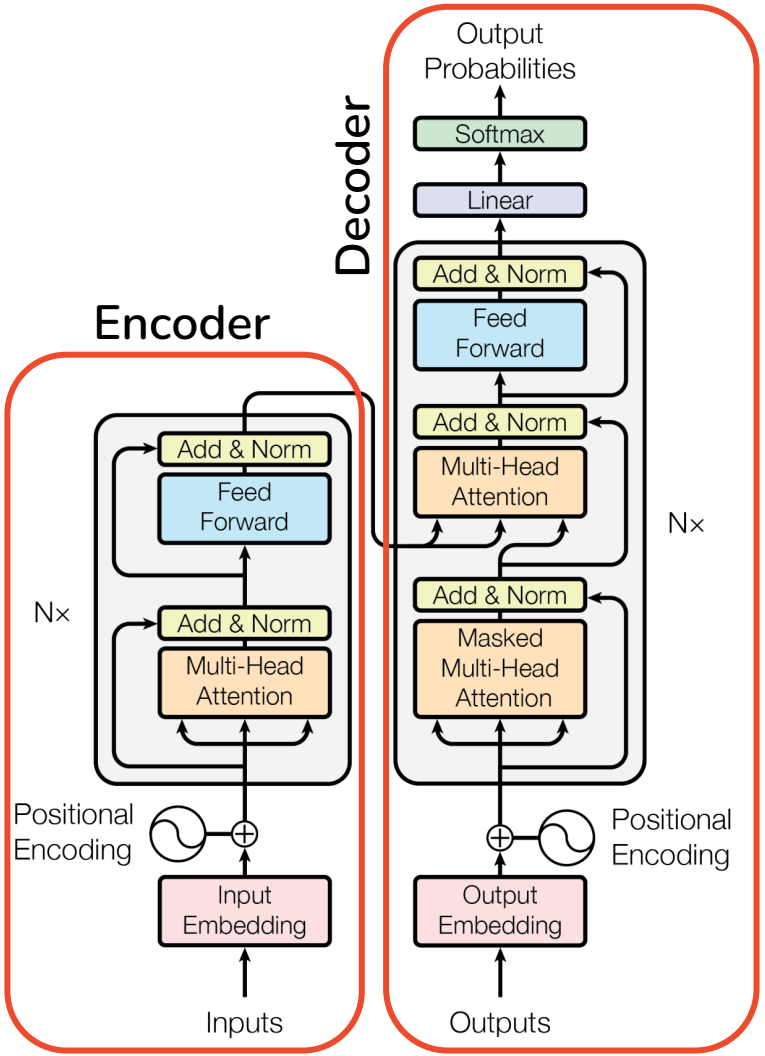
2. Feed-forward network: a 2-layer MLP (ReLU), applied to each position independently, to "process what attention gathered"

3. Residual connections: add a sublayer's input to its output, for easier gradient flow and deeper stacks

4. Layer normalization: per-token normalization, for stable training

# The Full Transformer

- Input → token embeddings + positional embeddings
- N transformer blocks → contextualized representations, layer by layer
- Output head → a final layer that maps representations to predictions
- ***Decoder:*** A transformer trained with masked attention learns to predict the next word, so it can generate text. This is the GPT family.
- ***Encoder:*** A transformer trained with bidirectional attention sees the whole sentence at once, so it is built to understand text. This is the BERT family.



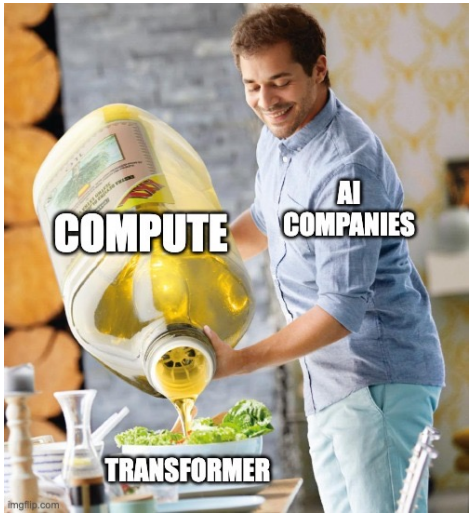
# The Success of the Transformer

| Model | BLEU | | Training Cost (FLOPs) | |
|---|---|---|---|---|
| | EN-DE | EN-FR | EN-DE | EN-FR |
| ByteNet [18] | 23.75 | | | |
| Deep-Att + PosUnk [39] | | 39.2 | | $1.0 \cdot 10^{20}$ |
| GNMT + RL [38] | 24.6 | 39.92 | $2.3 \cdot 10^{19}$ | $1.4 \cdot 10^{20}$ |
| ConvS2S [9] | 25.16 | 40.46 | $9.6 \cdot 10^{18}$ | $1.5 \cdot 10^{20}$ |
| MoE [32] | 26.03 | 40.56 | $2.0 \cdot 10^{19}$ | $1.2 \cdot 10^{20}$ |
| Deep-Att + PosUnk Ensemble [39] | | 40.4 | | $8.0 \cdot 10^{20}$ |
| GNMT + RL Ensemble [38] | 26.30 | 41.16 | $1.8 \cdot 10^{20}$ | $1.1 \cdot 10^{21}$ |
| ConvS2S Ensemble [9] | 26.36 | **41.29** | $7.7 \cdot 10^{19}$ | $1.2 \cdot 10^{21}$ |
| Transformer (base model) | 27.3 | 38.1 | **$3.3 \cdot 10^{18}$** | |
| Transformer (big) | **28.4** | **41.8** | $2.3 \cdot 10^{19}$ | |

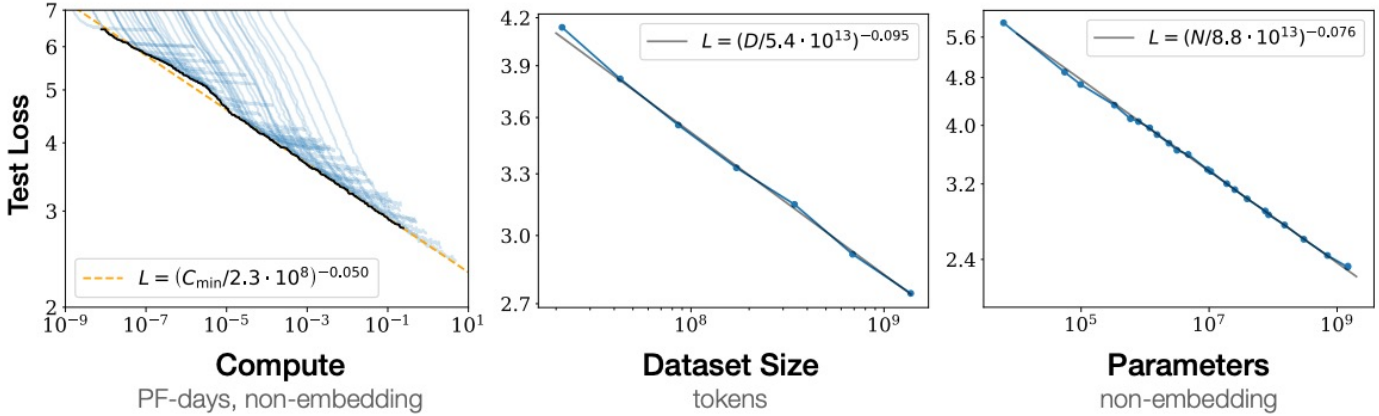
| | Base | Big |
|---|---|---|
| Layers in Encoder & Decoder | 6 | 6 |
| #Heads | 8 | 16 |
| Model Dimension | 512 | 1024 |
| Feed-Forward Dimension | 2048 | 4096 |

# The Success of the Transformer

- As it turns out, the Transformer architecture is amazing for pre-training models on huge datasets
- And the Encoder (BERT etc) or Decoder (GPT etc) themselves make great Language Models
- The rest is scaling (although attention has quadratic costs)

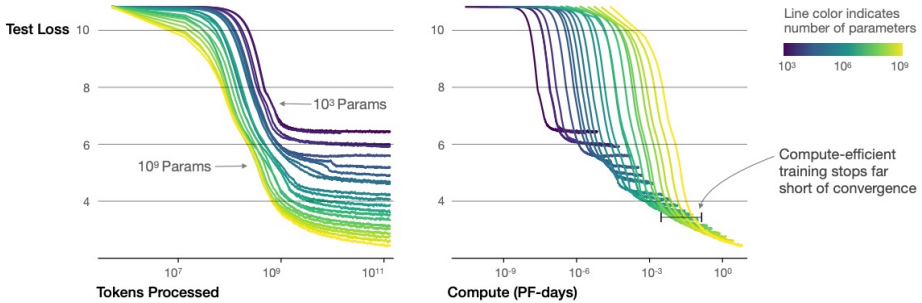


# Scaling Transformer Models



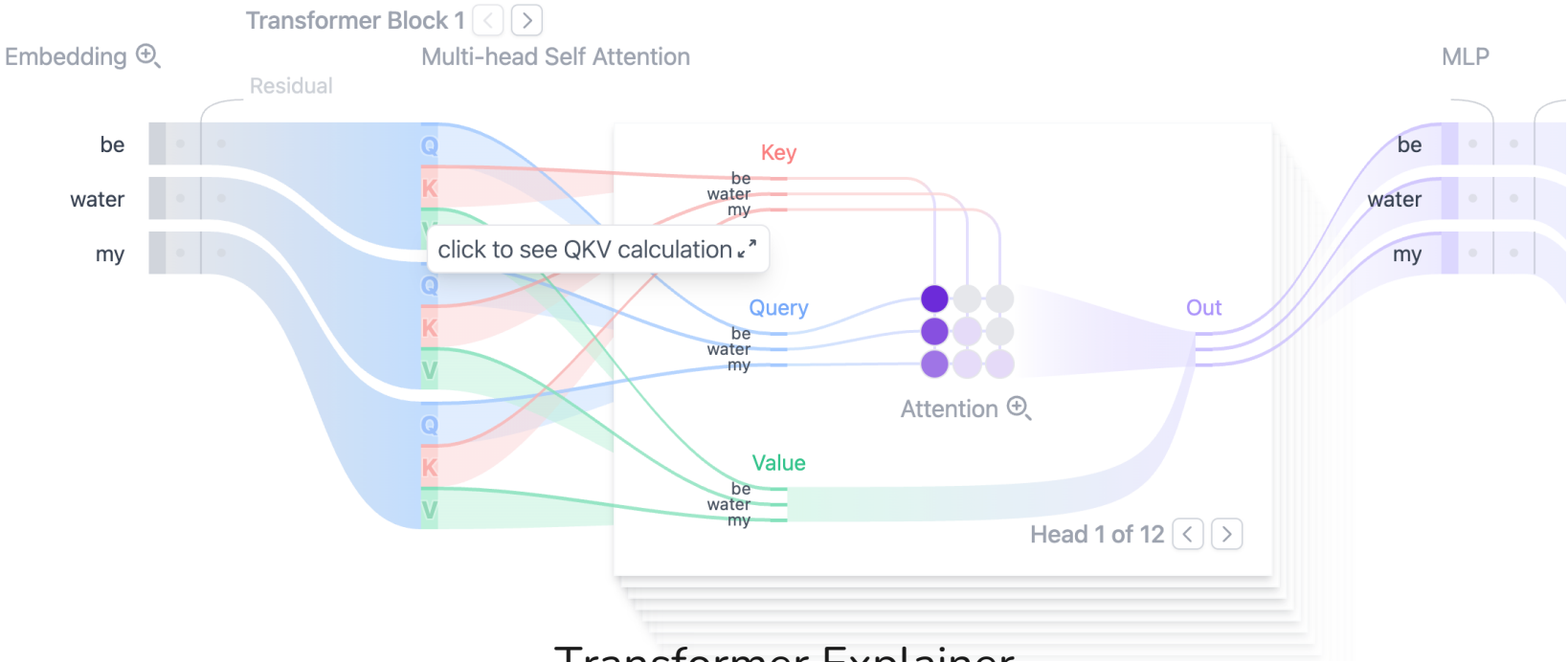

Larger models require **fewer samples** to reach the same performance

The optimal model size grows smoothly with the loss target and compute budget




Scaling Laws for Neural Language Models [OpenAI, 2020]

# Folks do great visualizations of the transformer




Transformer Explainer

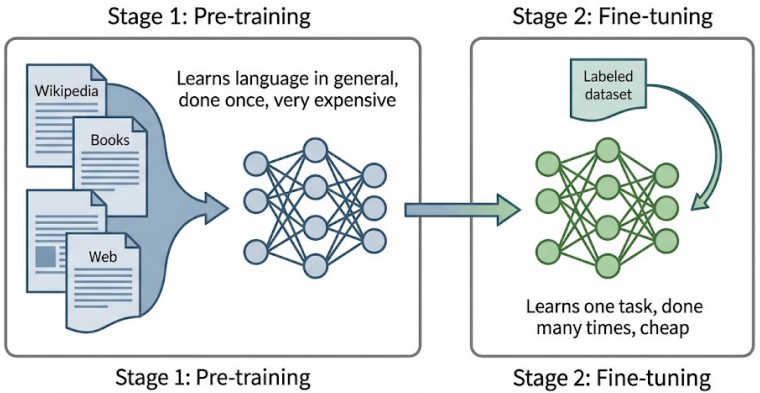
03.

# Pre-Training

# The Problem with Training from Scratch

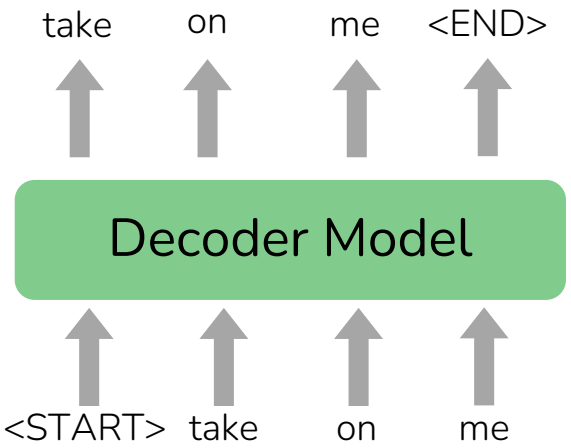
- So far, every task we have built starts almost from zero.
- Word vectors transfer, but the rest of the model has random parameters
- The model has no idea how language works before training begins: no grammar, no word senses, no world knowledge
- It must learn all of that plus the actual task, from a small labeled dataset

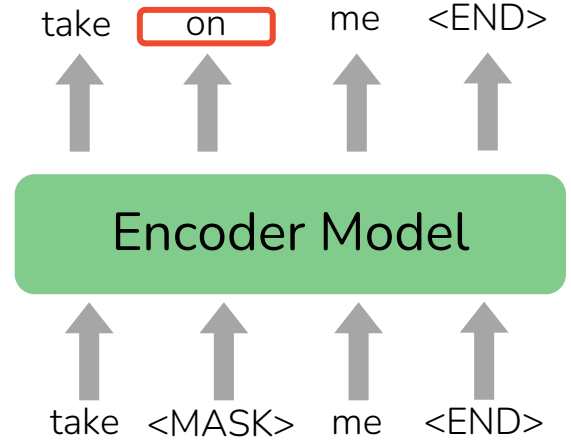
- Labeled data is expensive.
  Raw text is everywhere.
- The pre-training idea: first let a model learn language in general from huge amounts of raw text. Then teach it your specific task.



# Self-Supervision: Learning Without Labels

- Pre-Training a model works by masking some parts of the input and train the model to reconstruct these parts
- This allows us to process massive amounts of data without any labels (unsupervised)
- Two pre-training styles
- Masked: Pretraining through language modeling, i.e. masking the next word and let the model predict the next word
  → Decoder Models (GPT etc)
- Bidirectenional: Predit missing words in the middle like a cloze
  → Encoder Models (BERT etc)



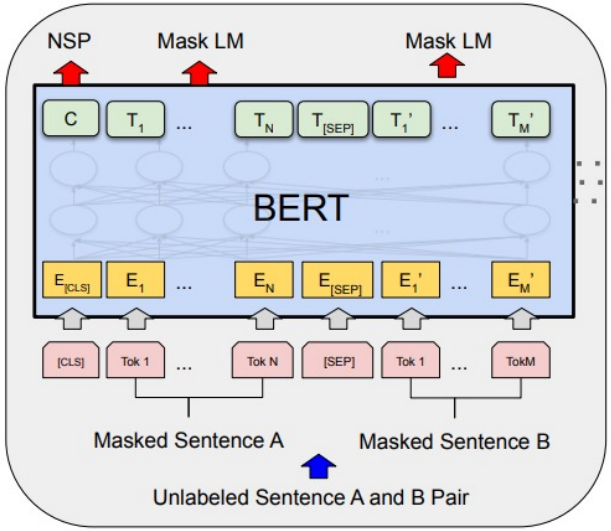



04.

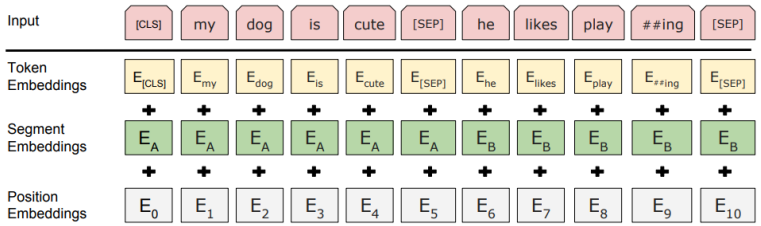
# BERT

# BERT: Bidirectional Encoder Representations from Transformers

- A stack of transformer blocks using bidirectional attention: every word sees the whole sentence (Encoder style) developed by Google
- Trained on Masked LM and ***next sentence prediction***
- Next sentence prediction: predict whether one sentence follows the other or randomly sampled
    - Other BERT variants neglect this
- Goal is not to generate text but to compute a rich representation of each word in context
- Pre-trained once on huge text, then fine-tuned for understanding tasks




Pre-training




BERT [Devlin et al 2018]

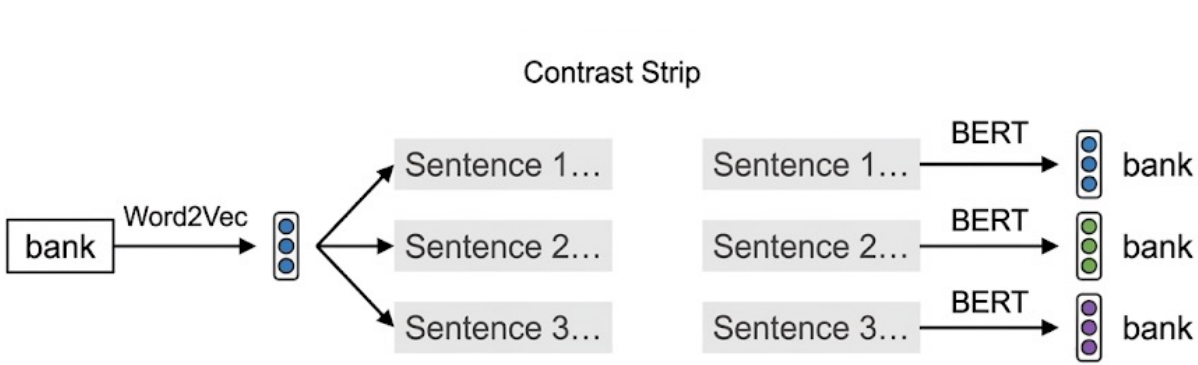
# BERT Training Details

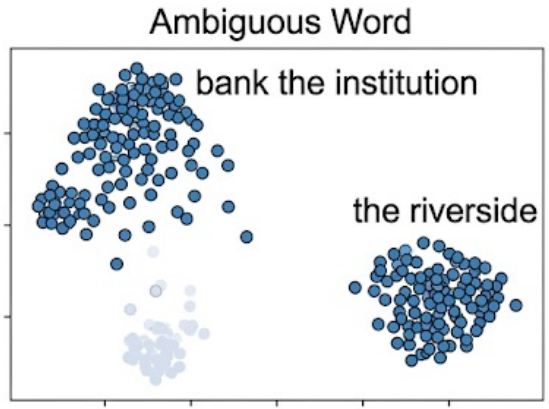
- Masked LM details: 15% of tokens are masked and out of these
  - 80% are replaced with [MASK]
  - 10% replaced with random other token
  - 10% are unchanged but still predicted
- BERT-base: 12 layers, 768-dim hidden states, 12 attention heads, 110M params.
- BERT-large: 24 layers, 1024-dim hidden states, 16 attention heads, 340M params.
- Trained on:
  - BooksCorpus (800 million words)
  - English Wikipedia (2,500 million words)
- BERT was pretrained with 64 TPU chips for a total of 4 days.

BERT [Devlin et al 2018]

# Where BERT's Word Vectors Come From

- BERT’s output vectors are contextual embeddings:
- For each input word, BERT's output vector $h_i$ is that word's meaning in this specific sentence
- The same word in a different sentence gets a different vector: one vector per word instance, not per word type
- These vectors are the result of running the sentence through the encoder, attention has mixed in the surrounding words layer by layer



# A Caveat: Raw BERT Vectors and Similarity

- It is tempting to compare two words (or two sentences) by the cosine similarity of their BERT vectors. This works poorly out of the box.
- Anisotropy: BERT's vectors nearly all point in similar directions. Two random words can have a cosine close to 1, so cosine no longer separates "related" from "unrelated" well.
- Caused partly by a few "rogue" dimensions with very large values that dominate the cosine
- Partial fix: standardize the vectors (subtract the mean, divide by the standard deviation)

See SentenceBERT & BERTTopic in the next lecture

05.

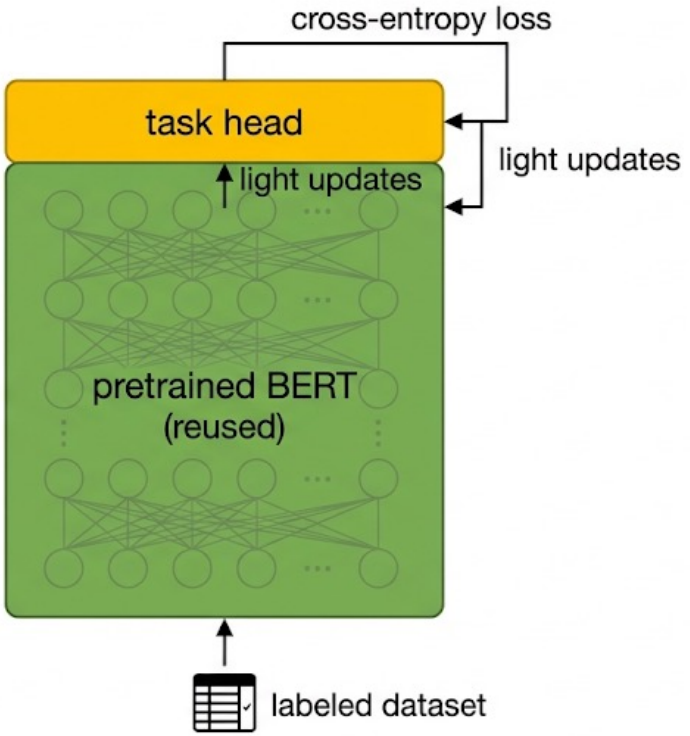
# Fine-Tuning BERT

# How Fine-Tuning Works

- The pretrained BERT already understands language. Fine-tuning just adapts it to a specific task.

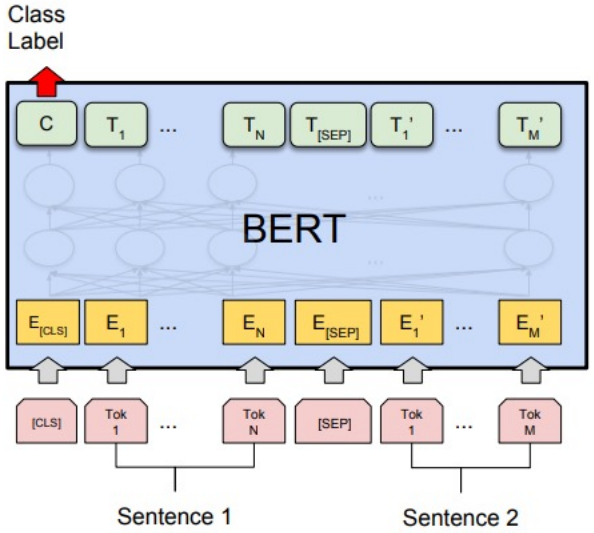
1. Take the pretrained BERT (downloaded, not trained from scratch)
2. Add a small task-specific head on top: usually a single new layer
3. Train on a labeled dataset for the task
4. Update the head, and lightly adjust BERT's own weights, using ordinary cross-entropy loss

- This is cheap: small dataset, few passes over the data, often only the top layers change much.

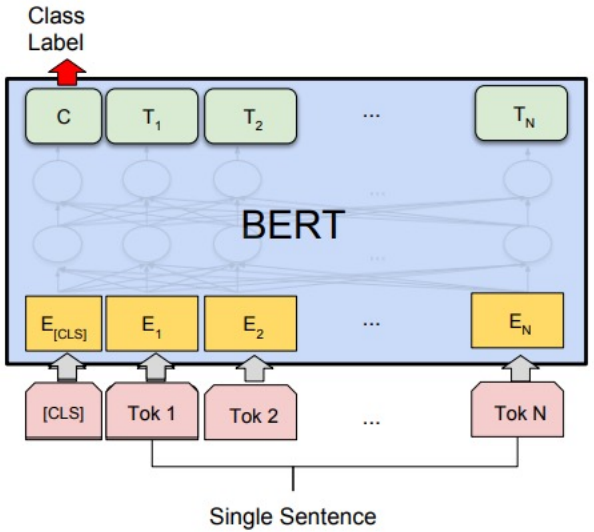


# BERT Fine-Tuning Task – Text Classification

- For Text Classification, we start every sentence with a special [CLS] token
- The context vector of this token can then be used in a linear layer and softmaxed

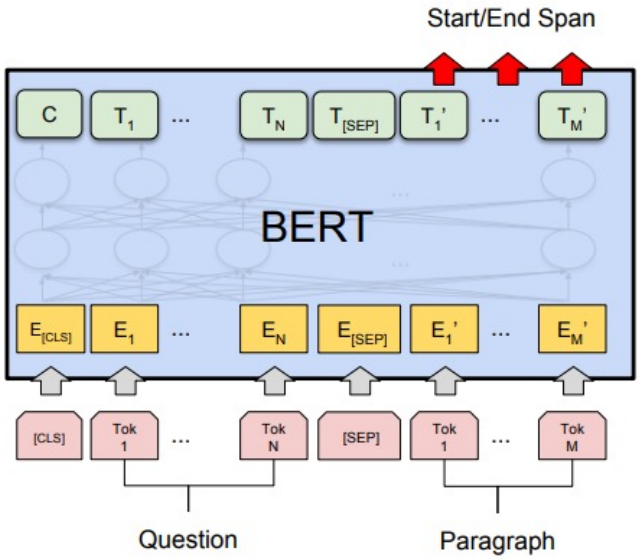



(a) Sentence Pair Classification Tasks: MNLI, QQP, QNLI, STS-B, MRPC, RTE, SWAG

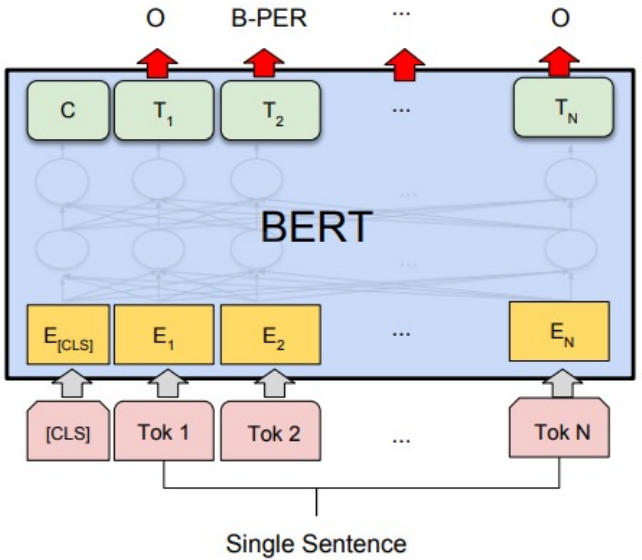



(b) Single Sentence Classification Tasks: SST-2, CoLA

# BERT Fine-Tuning Task – QA & NER Tagging




(c) Question Answering Tasks:
SQuAD v1.1




(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

# BERT Success

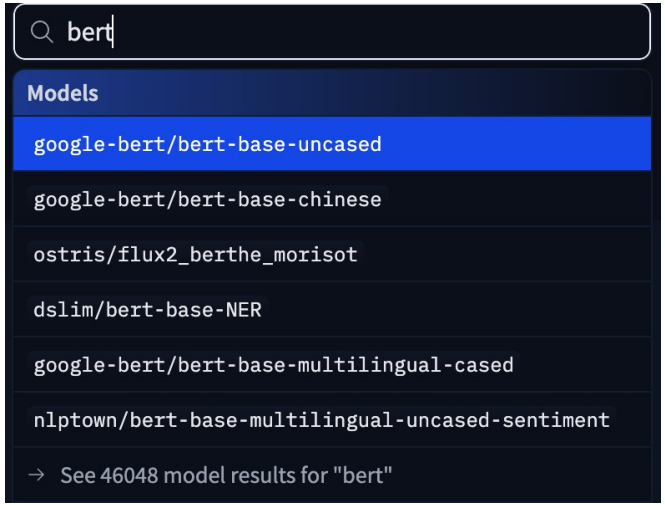
- By fine-tuning, BERT yielded state-of-the-art results on many tasks
- Is considered the powerhouse of modern NLP - in particular its variants
- Use it for any discriminative task in particular text classification

| System | MNLI-(m/mm) | QQP | QNLI | SST-2 | CoLA | STS-B | MRPC | RTE | **Average** |
|---|---|---|---|---|---|---|---|---|---|
| | 392k | 363k | 108k | 67k | 8.5k | 5.7k | 3.5k | 2.5k | - |
| Pre-OpenAI SOTA | 80.6/80.1 | 66.1 | 82.3 | 93.2 | 35.0 | 81.0 | 86.0 | 61.7 | 74.0 |
| BiLSTM+ELMo+Attn | 76.4/76.1 | 64.8 | 79.8 | 90.4 | 36.0 | 73.3 | 84.9 | 56.8 | 71.0 |
| OpenAI GPT | 82.1/81.4 | 70.3 | 87.4 | 91.3 | 45.4 | 80.0 | 82.3 | 56.0 | 75.1 |
| $\text{BERT}_{\text{BASE}}$ | 84.6/83.4 | 71.2 | 90.5 | 93.5 | 52.1 | 85.8 | 88.9 | 66.4 | 79.6 |
| $\text{BERT}_{\text{LARGE}}$ | **86.7/85.9** | **72.1** | **92.7** | **94.9** | **60.5** | **86.5** | **89.3** | **70.1** | **82.1** |

Table 1: GLUE Test results, scored by the evaluation server (https://gluebenchmark.com/leaderboard).

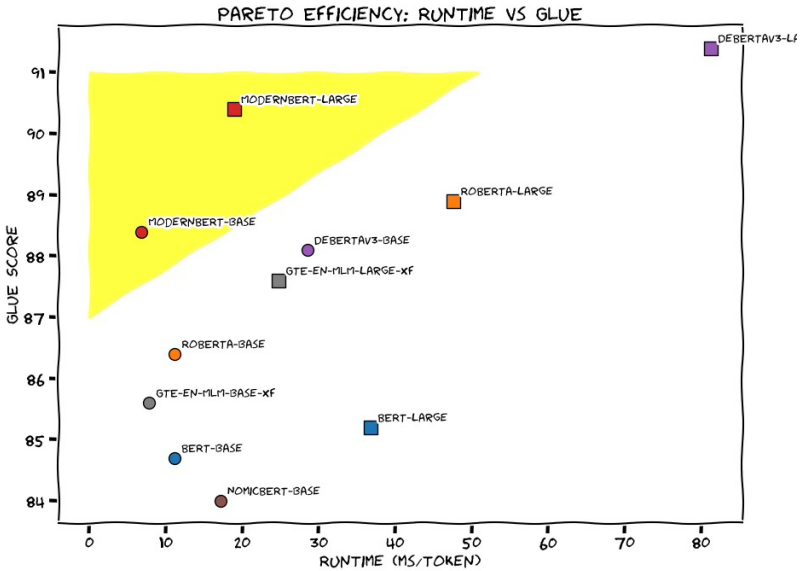
# BERT Model Family

- RoBERTa [2019]
  - No next sentence prediction
  - Dynamic masking (other masks every epoch)
  - BPE tokenizer instead of WordPiece
  - Small training adjustments
- DistilBERT
  - Much smaller version (40% less params) trained via student-teacher
- Lots of variants for different languages or use cases – just search HuggingFace 🤗
  - Sentiment analysis, emotion detection
  - German RoBERTa, multilingual variants (XLM-RoBERTa)



# modernBERT (12/2024)

- 8192 Tokens in the Context Window (vs 512 Vanilla BERT)
- Improved Transformer architecture using global and local Attention
- Lots of smaller tricks from years of LLM training experience
- EuroBERT (03/2025)
  - Basically, modernBERT trained on European languages




# The Hugging Face Ecosystem

- In practice, you rarely build or pre-train these models yourself. Hugging Face is the standard open platform for using them.
- The 🤗 Hub
  - A huge public repository of pretrained models (BERT and many more)
  - Thousands of ready-made datasets
  - Model cards describe each model's training, intended use, and limitations
- The key libraries
  - **`transformers`**: load any pretrained model and its tokenizer in a few lines
  - **`datasets`**: download and process datasets efficiently